

BFR Training Improves Patients' Reported Outcomes, Strength, and Range of Motion After Casting for Colles' Fracture

MINGMING YANG¹, BIN LIANG², XIN ZHAO², YANG WANG², MINGYUAN XUE³, and DAN WANG¹

¹*School of Elite Sport, Shanghai University of Sport, Shanghai, CHINA;* ²*Department of Orthopaedics, Nanjing First Hospital, Nanjing, CHINA;* and ³*Zimmer Biomet Inc., CHINA*

ABSTRACT

YANG, M., B. LIANG, X. ZHAO, Y. WANG, M. XUE, and D. WANG. BFR Training Improves Patients' Reported Outcomes, Strength, and Range of Motion After Casting for Colles' Fracture. *Med. Sci. Sports Exerc.*, Vol. 55, No. 11, pp. 1985–1994, 2023. **Purpose:** This study compared traditional rehabilitation as a treatment modality after plaster cast treatment of Colles' fracture with a combination of individualized blood flow restriction (BFR) and traditional rehabilitation. **Methods:** Twenty-eight participants were randomized into a BFR group and a non-BFR group after plaster cast treatment of Colles' fracture. The BFR group completed traditional rehabilitation with a medical grade tourniquet applied to the upper arm, and the non-BFR group underwent traditional rehabilitation only. Patients were followed up with radiographic outcomes (palmar tilt and radial inclination) to ensure the stability of the fracture. Clinical assessment of patient-rated wrist evaluation (PRWE) score, grip strength, pinch strength, wrist range of motion (ROM), and muscle stiffness was conducted at cast removal and 6 wk after cast removal. Two-way repeated-measures ANOVA determined significant interactions between time and group in the aforementioned variables. An independent-sample *t*-test assessed the differences in baseline variables and radiographic outcomes. **Results:** Significant interactions between time and group were noted for PRWE score ($F = 11.796$, $P = 0.002$, $\eta_p^2 = 0.339$), grip strength ($F = 5.445$, $P = 0.029$, $\eta_p^2 = 0.191$), and wrist ROM (ulnar deviation; $F = 7.856$, $P = 0.010$, $\eta_p^2 = 0.255$). No significant interactions between time and group were found in measurements of pinch strength or wrist ROM (flexion, extension, radial deviation, pronation, supination). An independent-sample *t*-test showed no significant difference in baseline variables and radiographic outcomes between the groups before or after intervention. **Conclusions:** This study found that combining individualized BFR with traditional rehabilitation resulted in greater increases in PRWE score, grip strength, and wrist ROM (ulnar deviation) than traditional rehabilitation alone. Therefore, adding individualized BFR to traditional rehabilitation might be a better option for treatment for similar patients. **Key Words:** VASCULAR OCCLUSION TRAINING, POSTTREATMENT REHAB, TRADITIONAL REHABILITATION TRAINING, BLOOD FLOW RESTRICTION TRAINING

Distal radius fracture (DRF) is one of the most common types of fracture, especially in people older than 50 yr (1). Colles' fracture is the most common type of DRF and is the predominant reason for reduced independence and long-lasting disability in Colles' fracture patients (2). Plaster cast is the most frequent treatment method for patients with Colles' fractures. Studies have indicated that patients experience decreased grip and pinch strength, loss of range of motion (ROM) in the wrist, increased muscle stiffness, and hand disability after Colles' fracture (3). In fact, 15% of patients sustaining a Colles' fracture report long-term pain and disability after

20 yr (4). Therefore, a timely and effective rehabilitation program conducted at an early stage after a Colles' fracture is critical, as the greatest amount of recovery occurs within the first 3 to 6 months after the fracture (5).

Low-load resistance training is the traditional method of rehabilitation for patients with Colles' fractures after the removal of a plaster cast (6). However, it recruits fewer type II fibers when compared with high-load resistance training, which would limit muscle hypertrophy (7). Blood flow restriction (BFR) training is a popular rehabilitation method for some musculoskeletal injuries (including knee arthroscopy, total knee arthroplasty, and Achilles tendon ruptures) (8). This type of training is often a complete occlusion of venous return and partial restriction of arterial inflow while performing low-intensity strength training to induce muscle hypertrophy and improve strength, achieving similar effects to those of high-intensity strength training (9). BFR training initially gained popularity in athletes and was found to be safe for inducing muscle hypertrophy without maximal or submaximal effort (10). Because patients were unable to undergo high-intensity resistance training in the early stages after injury, BFR training emerged as a promising method to mimic those effects during rehabilitation. It has

Address for correspondence: Dan Wang, Ph.D., Shanghai University of Sport, 399 Changhai Road, Shanghai, China; E-mail: wangdan@sus.edu.cn.

M. Y. and B. L. contributed equally to the work.

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been shown to be effective in promoting postoperative muscle recovery after knee arthroscopy, total knee arthroplasty, and Achilles tendon ruptures (8). Therefore, low-load (or no load) BFR is suitable for this population, given its effects of improved skeletal muscle maintenance, decreased mechanical stress on the joints, and decreased muscle damage after training (11). BFR training was found to result in better pain reduction during activity and greater improvement in overall self-perceived function compared with the traditional rehabilitation protocol in study participants with nonoperative DRF (12). One study has shown that people presented different tolerances and neuromuscular fatigue in response to different training loads, which affected the effectiveness of rehabilitation training program (13). Of note, the training load for the rehabilitation training program in this study was not individualized. Also, the age range of the patients included in this study was 18 to 65 yr. Another study showed that people of different ages have different abilities to adapt to muscle hypertrophy during strength training (14). Therefore, the large age span of participants may have led to bias about the effectiveness of this rehabilitation.

It was hypothesized that individualized BFR training combined with traditional rehabilitation training would produce better recovery (measured by patient-rated wrist evaluation (PRWE) score, grip strength, pinch strength (tip pinch strength, key pinch strength, and tripod pinch strength), wrist ROM, brachioradialis, and abductor pollicis brevis stiffness) for participants older than 50 yr who had undergone plaster cast treatment of Colles' fracture.

METHODS

Participants. G*Power Version 3.1.9.7 was used to calculate the required sample size. Because no similar published data were found to work out the sample size, sample size was calculated according to medium and large effect size. Based on a partial η^2 of 0.14 (large effect), effect size was calculated as 0.4. According to the effect size of 0.4, a sample size of 16

was determined. Based on a partial η^2 of 0.06 (medium effect), effect size was calculated as 0.25. According to the effect size of 0.25, a sample size of 34 was determined. Thirty-four participants were a conservative estimate to detect a statistically significant result in this study. We called 72 potential patients to inform the context of the study. Because of the COVID-19 outbreak, only 28 people were willing to participate in our study. The sample size of 28 was within the range of our calculated sample size: 16–34. Therefore, 28 patients scheduled for plaster cast treatment of Colles' fracture were eventually recruited for this study. After treatment, participants were stratified sampling based on sex and randomized to the BFR or non-BFR groups using a computer-generated randomization program. Inclusion criteria for the participants were as follows: 1) Colles' fracture, 2) age between 50 and 75 yr (Colles' fracture is most commonly diagnosed in people older than 50 yr, and people older than 75 yr have poor cardiovascular and cognitive function), 3) no record of wrist injury or joint deformity, 4) no other types of upper limb injuries occurred in the last 3 months, and 5) no cardiovascular diseases. Exclusion criteria of the participants were as follows: 1) concurrent ipsilateral upper limb fracture, 2) concurrent bilateral upper limb fractures, 3) record of surgery or any invasive procedure on the affected upper limb, 4) record of peripheral arterial disease or deep vein thrombosis, and 5) record of cancer that has resulted in limitations or restrictions to physical exercise. Each enrolled participant signed an informed consent form. Three participants were lost before completing the study protocol (1 in the BFR group and 2 in the non-BFR group) because of the COVID-19 outbreak, leaving 25 participants completed all sessions. No adverse events were reported. The baseline variables and demographic characteristics of the BFR and non-BFR groups before training did not significantly differ from each other (Table 1). The ethical committee of the Nanjing First Hospital approved the study before initiation of testing (KY20200820-01). This clinical trial was registered on ClinicalTrials.gov (ID: NCT05371431).

TABLE 1. Demographic characteristics and baseline variables in the BFR group and non-BFR group.

	BFR Group (<i>n</i> = 13)		Non-BFR Group (<i>n</i> = 12)		<i>P</i>
	Male (<i>n</i> = 2)	Female (<i>n</i> = 11)	Male (<i>n</i> = 2)	Female (<i>n</i> = 10)	
LOP	154.0 ± 7.07	131.55 ± 11.18	—	—	—
Age (yr)	61.5 ± 9.2	64.7 ± 7.3	56.0 ± 1.4	64.1 ± 4.8	0.482
Height (cm)	171.0 ± 1.4	159.6 ± 4	167.0 ± 2.8	159.7 ± 5.8	0.821
Preintervention weight (kg)	70.50 ± 3.54	59.64 ± 8.99	67.0 ± 9.90	58.90 ± 6.40	0.905
Postintervention weight (kg)	70.00 ± 7.07*	60.36 ± 9.02*	68.05 ± 9.26*	59.67 ± 6.51*	0.957
PRWE		135.85 ± 13.28		126.42 ± 14.41	0.631
Grip strength		5.05 ± 3.85		4.16 ± 3.10	0.788
Tip pinch		1.94 ± 0.97		1.69 ± 1.26	0.258
Key pinch		2.98 ± 1.27		2.79 ± 1.84	0.299
Tripod pinch		2.15 ± 1.13		2.08 ± 1.67	0.089
Wrist flexion		26.50 ± 7.16		24.78 ± 6.83	0.493
Wrist extension		20.32 ± 13.51		16.60 ± 9.53	0.205
Ulnar deviation		15.38 ± 4.85		14.78 ± 5.19	0.730
Radial deviation		6.71 ± 2.81		8.43 ± 3.07	0.582
Pronation		50.48 ± 16.39		39.88 ± 13.04	0.718
Supination		45.64 ± 21.45		36.98 ± 23.97	0.834
Brachioradialis stiffness		244.77 ± 28.92		270.50 ± 37.53	0.331
Abductor pollicis brevis stiffness		394.92 ± 54.15		398.75 ± 54.98	0.815
Radial inclination		21.31 ± 2.43		22.08 ± 2.68	0.915
Palmar tilt		12.23 ± 1.59		13.67 ± 1.61	0.981

*No significant difference compared with preintervention weight ($P > 0.05$).

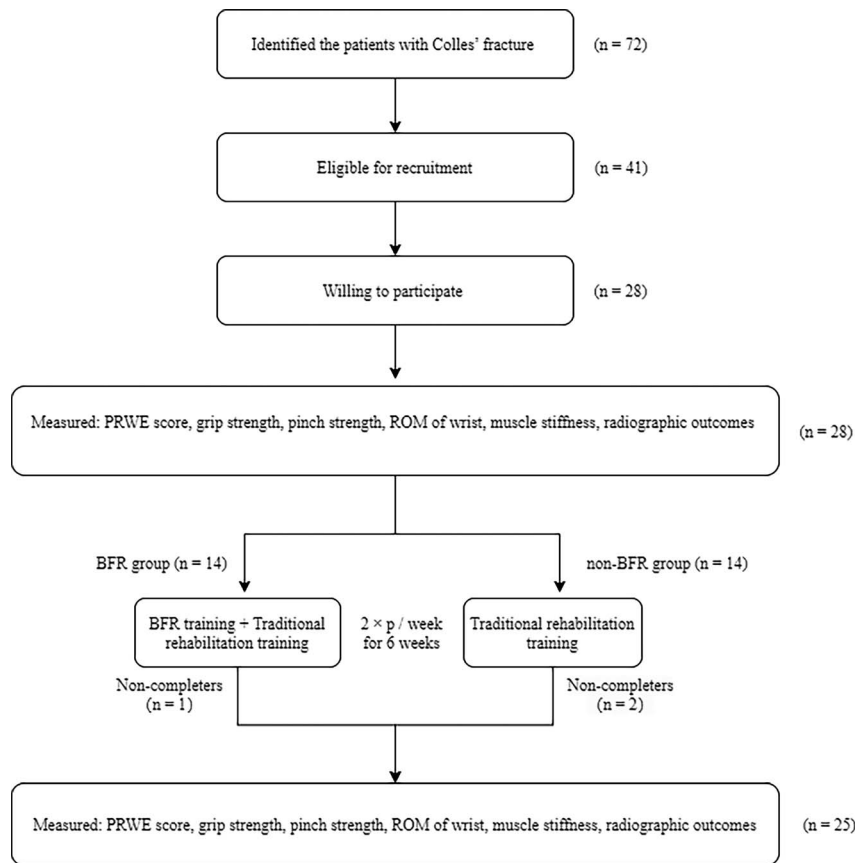


FIGURE 1—Experimental design and study procedure.

Experimental design and procedure. The experimental design and procedure of the study are shown in Figure 1. Traditional rehabilitation training (Table 2) was designed based on postfracture rehabilitation guidelines (15). This guideline recommends that rehabilitation of early-stage DRF fractures should be progressed gradually, with training loads ranging from self-weight to loads differentiated based on the patient's individual situation. Participants attended training sessions twice a week for 6 wk with at least 48 h of rest between sessions. Individualized BFR training (twice a week for at least 6 wk) has been reported to increase muscle hypertrophy and strength (16). Participants had an initial load of 35% one-repetition maximum for 12 repetitions corresponding to rating of perceived exertion score of 11 on the Borg 6–20 scale (17). For the rest of the training session, the training load was increased by 1 lb (~0.45 kg) if the participants could complete four sets, 75 (30, 15, 15, 15 repetitions) repetitions

per set in the two training sessions during the previous training week (18). Otherwise, the participants would maintain the training load from the previous week.

BFR procedure. The BFR group underwent traditional rehabilitation training combined with a medical-grade tourniquet system (A.T.S.® 4000TS Tourniquet System; Zimmer Surgical, Inc., Dover, OH). The tourniquet system consists of an A.T.S.® 4000TS unit and a tourniquet cuff (10 × 46 cm) with protection sleeve. One study discovered that a wide cuff width (10 cm) resulted in less pain and greater safety for patients and has a significant effect on increasing muscle strength (19). The A.T.S.® 4000TS Tourniquet System tailors the tourniquet pressure to each participant after determination of limb occlusion pressure (LOP). A study has shown that 50% LOP is safe and effective in DRF rehabilitation (12). LOP values were reassessed before each training session with the tourniquet system, and pressures were continually monitored during the sessions.

TABLE 2. Traditional rehabilitation training program for participants enrolled in this study.

Exercises	Load	Frequency	Volume	Intrasets Rest (s)	Duration (wk)
Grip and pinch	Individualization	2 per week	4 sets of 30/15/15/15 repetitions	45	6
Wrist flexion with pronated forearm	Individualization	2 per week	4 sets of 30/15/15/15 repetitions	45	6
Wrist extension with pronated forearm	Individualization	2 per week	4 sets of 30/15/15/15 repetitions	45	6
Wrist flexion with supinated forearm	Individualization	2 per week	4 sets of 30/15/15/15 repetitions	45	6
Wrist extension with supinated forearm	Individualization	2 per week	4 sets of 30/15/15/15 repetitions	45	6
Prayer sign (wrist flexion)	Individualization	2 per week	4 sets of 30/15/15/15 repetitions	45	6
Prayer sign (wrist extension)	Individualization	2 per week	4 sets of 30/15/15/15 repetitions	45	6
Forearm pronation	Individualization	2 per week	4 sets of 30/15/15/15 repetitions	45	6
Forearm supination	Individualization	2 per week	4 sets of 30/15/15/15 repetitions	45	6

2. Abductor pollicis brevis: The muscle contraction was confirmed by extending the thumb radially.

The probe of the myometer was perpendicularly lowered onto the muscle belly, and an automatic mechanical impact was delivered to the muscle (duration of 15 or 12 ms with a force of 0.3–0.4 N, and a local deformation in the order of a few millimeters). Resultant damped natural oscillations were measured using a built-in accelerometer sampling at 3200 Hz (26). Five consecutive measurements were taken, and the average of the five measurements was used for later analysis.

Radiographic outcomes measurement. Radial inclination and palmar tilt have been determined to be the most important radiographic variables for evaluating the successful reduction of Colles' fracture (27). Frontal and lateral radiographs were taken using digital radiology systems (DRX-Evolution Plus System; Carestream Health, Rochester, NY). Radial inclination (normal range, 21°–25°) is the angle between the line drawn from the tip of the radial styloid to the medial corner of the articular surface of the radius and the line drawn perpendicular to the long axis of the radius (28). Palmar tilt (normal range, 10°–15°) is the angle between the line connecting the most distal points of the volar and dorsal lips of the radius on a true lateral view and the longitudinal axis of the radius (28). Radiographic outcomes (radial inclination and palmar tilt) were measured by the same doctor as safety indicators.

Statistical analysis. All statistical analyses were performed using IBM SPSS Statistics for Windows, Version 26.0 (IBM Corp., Armonk, NY). Normal distribution of the data was tested using the independent-samples Shapiro–Wilk test. Homogeneity of variance was verified by Levene's test. PRWE score, grip strength, pinch strength (tip pinch, key pinch, and tripod pinch), wrist ROM, and muscle stiffness were each assessed using a 2 × 2 (group × time) repeated-measures ANOVA with group allocation (BFR vs non-BFR) as the between-participant independent factor, and time (preintervention and postintervention) as the within-participant dependent factor. An independent-sample *t*-test assessed the differences in radiographic outcomes and baseline variable. The interpretation of the effect size was as follows: partial eta-squared (η_p^2): small ≥ 0.01 , medium ≥ 0.06 , and

large ≥ 0.14 ; and Cohen's *d*: small ≥ 0.2 , medium ≥ 0.5 , and large ≥ 0.8 (29). Alpha significance was set *a priori* at $P < 0.05$. For each measurement, trials were recorded subsequently, and the mean and coefficient of variation (CV) were calculated. The CV was calculated by dividing the SD by the mean (AVE) of the trials and multiplying by 100, using the formula: $CV = (SD/AVE) \times 100$. For the grip strength measurement, the CV ranging from 7.5% to 20% on injured side is claimed to possess excellent reliability (30). For ROM of wrist, the CV ranging from 5.1% to 19.0% on injured side is reliable (31).

RESULTS

PRWE score. A two-way repeated-measures ANOVA found a significant interaction between time and group in the PRWE score ($F = 25.452$, $P < 0.001$, $\eta_p^2 = 0.525$; Fig. 2).

Muscle strength. For grip strength and pinch strength, the CV values were 10.98% and 5.76%, respectively. The CV values were within the cutoff value, suggesting that all test results were accurate and reliable. As shown in Figure 3, a two-way repeated-measures ANOVA found a significant interaction between time and group in grip strength ($F = 5.445$, $P = 0.029$, $\eta_p^2 = 0.191$). No significant interactions between time and group were found in key (lateral) pinch strength ($F = 0.019$, $P = 0.892$, $\eta_p^2 = 0.001$), tripod pinch strength ($F = 0.079$, $P = 0.781$, $\eta_p^2 = 0.003$), and tip pinch strength ($F = 0.000$, $P = 0.999$, $\eta_p^2 = 0.000$).

Wrist ROM. For wrist ROM, the CV value was 11.46%. The CV values were within the cutoff value, suggesting that all test results were reliable. Two-way repeated-measures ANOVA found a significant interaction between time and group in terms of ulnar deviation ($F = 7.856$, $P = 0.010$, $\eta_p^2 = 0.255$). No significant interactions between time and group were found in flexion ($F = 1.78$, $P = 0.195$, $\eta_p^2 = 0.072$), extension ($F = 1.284$, $P = 0.269$, $\eta_p^2 = 0.053$), radial deviation ($F = 3.044$, $P = 0.094$, $\eta_p^2 = 0.117$), pronation ($F = 1.286$, $P = 0.269$, $\eta_p^2 = 0.053$), and supination ($F = 0.299$, $P = 0.59$, $\eta_p^2 = 0.013$; Fig. 4).

Muscle stiffness. The CV value was 8.29% for muscle stiffness measurement. Two-way repeated-measures ANOVA

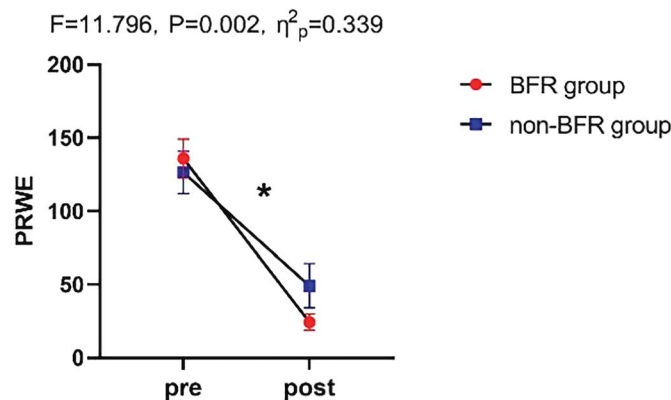


FIGURE 2—Comparison of PRWE scores before and after intervention. *Significant difference between the BFR and non-BFR groups from preintervention to postintervention.

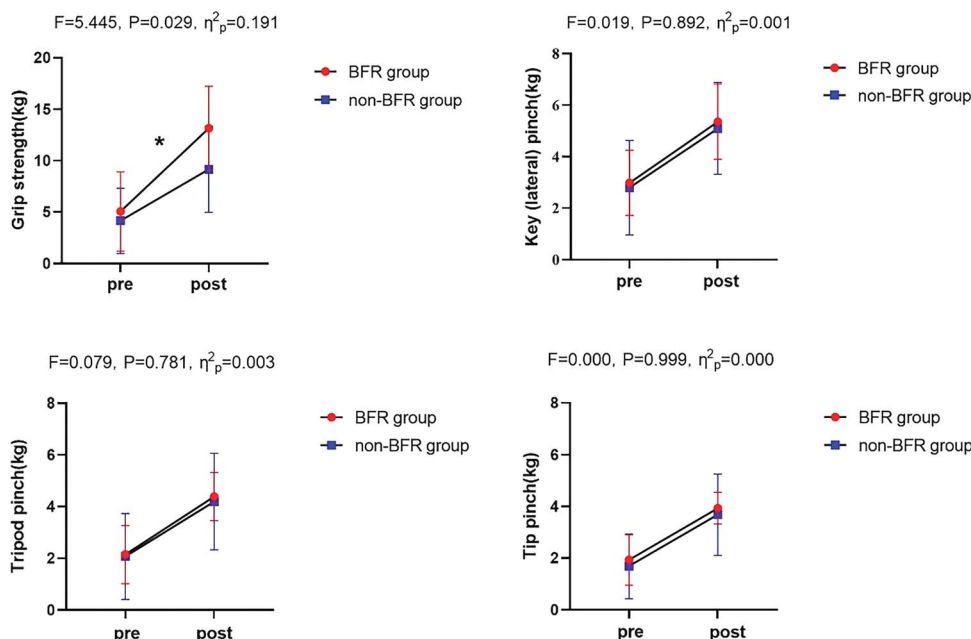


FIGURE 3—Comparison of muscle strength before and after intervention. *Significant difference between the BFR and non-BFR groups from preintervention to postintervention.

found no significant interactions between time and group in brachioradialis stiffness ($F = 1.56$, $P = 0.224$, $\eta^2_p = 0.064$) and abductor pollicis brevis stiffness ($F = 0.022$, $P = 0.883$, $\eta^2_p = 0.001$; Fig. 5).

Radiographic outcomes. An independent-sample *t*-test showed no significant differences in palmar tilt ($t = -2.240$; 95% confidence interval, -2.76 to -1.09 ; $P = 0.981$; $d = 0.90$) and radial inclination ($t = -0.759$; 95% confidence interval, -2.89 to -1.34 ; $P = 0.915$; $d = 0.30$) before and after intervention (Fig. 6).

DISCUSSION

This study discovered that, compared with traditional rehabilitation, individualized BFR combined with traditional rehabilitation after plaster cast treatment of Colles' fracture may result in a larger increase in grip strength, wrist ROM (ulnar deviation), and PRWE score. The Colles' fractures recovered satisfactorily according to radiographic variables during individualized BFR training.

PRWE score. This study found that, compared with traditional rehabilitation, individualized BFR training combined with traditional rehabilitation led to a significantly greater decrease in PRWE scores in participants with Colles' fracture. The PRWE score consists of a series of questions related to a patient's pain and the function of the wrist (specific activities and daily activities). Recent research has suggested that pain was significantly reduced after individualized BFR training compared with low-load resistance training in patients with DRF (12). Ischemia and pressure-induced muscle pain are often used as conditioned stimuli for pain modulation and have been shown to alter pain sensitivity in healthy individuals (32). Training-induced hypoalgesia (TIH) describes an acute

reduction in pain sensitivity after training (33). The intensity of training was acknowledged as an important determinant of the magnitude of TIH (34). One study provided evidence of greater TIH induced by low-intensity BFR resistance training compared with low-intensity resistance training (32). Among the mechanisms of conditioned pain modulation, BFR stimulation may act as a modulating stimulus, that is, pain and discomfort caused by BFR training suppresses pain associated with the fracture (32). During BFR resistance training, discomfort and pain sensations were exacerbated to a similar or greater extent than during high-intensity resistance training (35). This was attributed to the stimulation of group III and IV afferent fibers and the subsequent increase in sympathetic activity. The sympathetic nervous system acts to depress pain perception via the descending inhibition of nociception through sympathetic cell bodies in the intermediolateral tracts of the spinal cord (36). Therefore, the pain levels of participants in the BFR group decreased. The reduction in pain assisted participants in manual dexterity exercises and activities of daily living (such as the ability to cut food, dress, wash, and do housework), and thus significantly improved their PRWE scores.

Muscle strength. This study noted that, compared with traditional rehabilitation, individualized BFR training combined with traditional rehabilitation induced a larger increase in grip strength in participants with Colles' fracture who had undergone plaster cast treatment. The result was consistent with the findings of Bowman et al. (37), who observed increased grip strength in the BFR group when compared with the non-BFR group. First, BFR is known to induce metabolic accumulations, which may include biomarkers such as whole blood lactate and muscle cell lactate (38). The accumulation of these metabolites produces an acidic intramuscular environment.

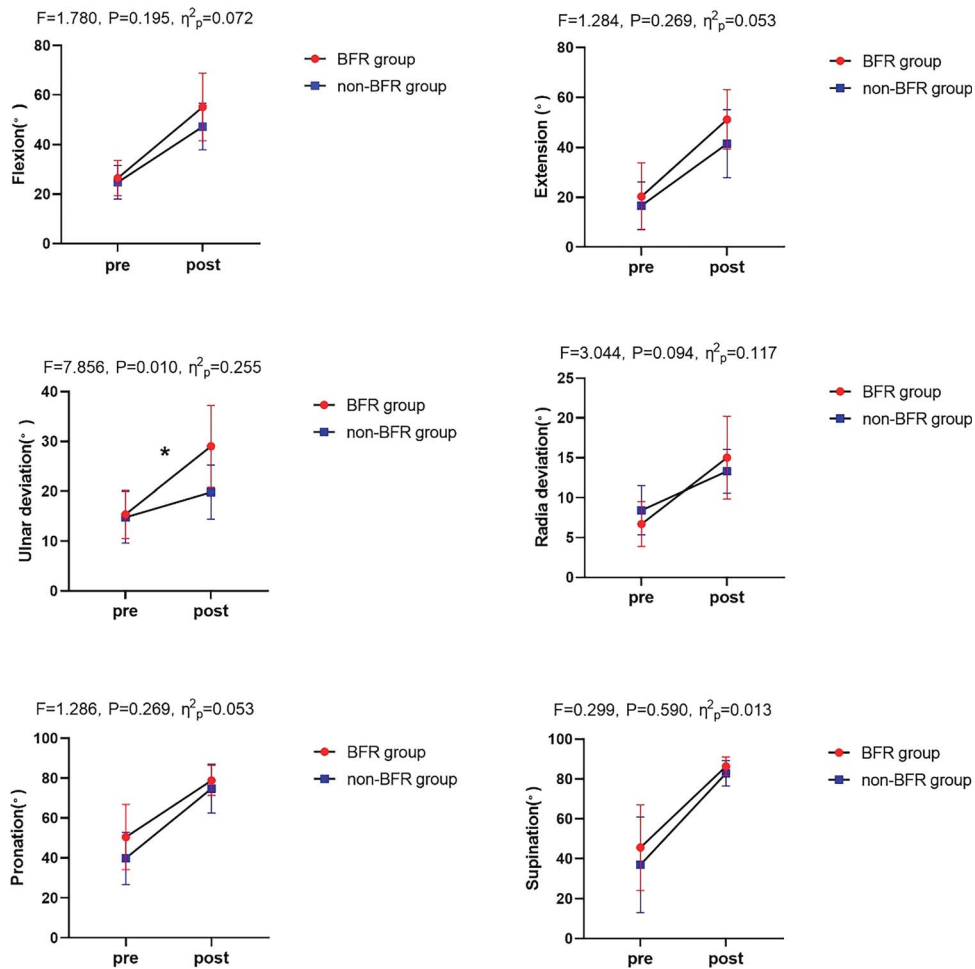


FIGURE 4—Comparison of wrist ROM before and after intervention. *Significant difference between the BFR and non-BFR groups from preintervention to postintervention.

The acidic muscle environment stimulates sympathetic nerve activity, which could promote the recruitment of more muscle fibers (38). Second, the pressure cuff helps to create a hypoxic environment for the muscle fibers, prompting an increase in angiogenic factors such as vascular endothelial growth factor, hypoxia inducible factor-1, and nitric oxide synthase (39). These angiogenic factors contribute to the anabolic effects in muscle growth (39). Therefore, the grip strength of the BFR group was significantly improved. However, there were no significant interactions between time and group in pinch strength (tip, key, and tripod). Similar results were also found in the study by Cancio et al. (12), in which pinch strength (tip,

key, and tripod) did not change significantly after BFR training when compared with traditional rehabilitation. Pinch strength corresponds to the patient's ability to complete fine movements and involves smaller muscle groups (such as the flexor pollicis longus) (40). Manini et al. (41) discovered that, during BFR training, muscle fibers were selectively recruited; that is, BFR training activates larger muscle groups first. This may explain the nonsignificant interactions between time and group in pinch strength.

Wrist ROM. The present study did not find significant interactions between time and group for ROM of wrist flexion, extension, radial deviation, forearm supination, and pronation,

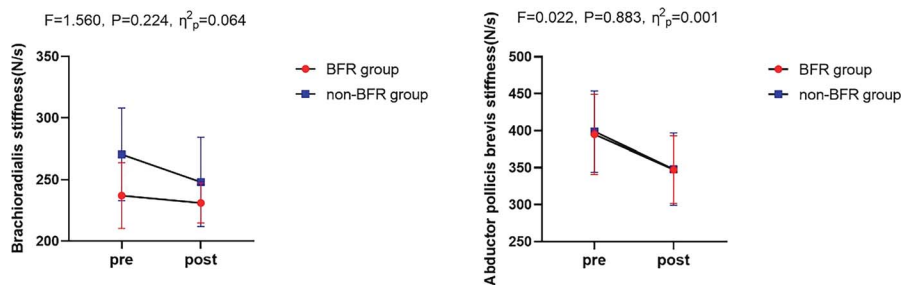


FIGURE 5—Comparison of brachioradialis and abductor pollicis brevis stiffness before and after intervention.

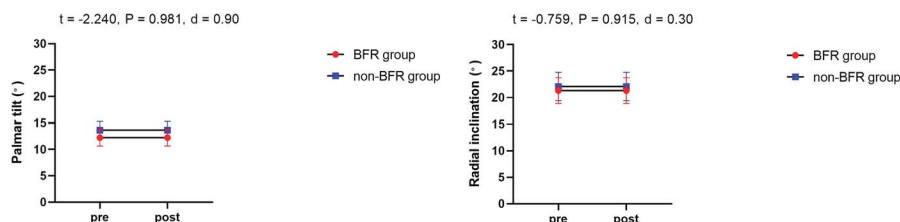


FIGURE 6—Radiographic outcomes were measured at preintervention and postintervention.

although they improved in both groups after the 6-wk intervention. A significant increase in ulnar deviation was discovered after BFR combined with traditional rehabilitation when compared with the traditional rehabilitation group. Colles' fractures of the participants in this study were fixed in the position of palmar flexion by plaster cast, which is a common position for fractures of this kind (42). One study showed that Colles' fracture immobilization in palmar flexion provided considerable restriction on ROM of ulnar deviation (42). The literature related to the influence of BFR training on ROM changes is limited. Studies have shown that BFR training may have a hypoalgesia effect on wrist (12), which could be one of the reasons for improved wrist ROM in this study. Therefore, ulnar deviation changed the most after intervention and showed a greater change in the BFR group. The restoration of ROM can be determined by the level of pain, swelling, damage to the soft tissues (muscle shortening, adhesions), and stiffness of wrist, among others; therefore, a multimodal intervention could be more suitable for recovery of ROM (43). This might explain the nonsignificant results in wrist ROM, because only strength training accompanied by active wrist ROM exercises was included in the intervention. Studies have shown that passive training can significantly improve wrist ROM with manual passive stretching (44). Also, physical therapy, such as paraffin bath therapy, relaxes the smooth muscle fibers in the small arteries, leading to peripheral vasodilation, which effectively improves blood circulation (45). In the future, a combination of physiotherapy treatment and passive training should be considered in individualized BFR training studies.

Muscle stiffness. This study found no significant interactions between time and group in brachioradialis and abductor pollicis brevis stiffness; however, there was a significant decrease in the stiffness value in both groups. Muscle stiffness was associated with swelling, soft tissue damage (muscle shortening, adhesions), and restricted venous blood flow (43). As previously mentioned, improved vascular endothelial function and peripheral blood circulation induced by BFR training could also be the reason for decreases in brachioradialis and abductor pollicis brevis stiffness (46). It has a regulatory effect on vasomotion and is thus beneficial to venous blood flow (46). Research has shown that traditional rehabilitation (e.g., strength training) can accelerate the blood flow of veins and lymphatic tissues, which is conducive to promoting muscle contraction (47). Muscle contraction reduces swelling, whereas reduced

swelling helps to decrease muscle stiffness (43). Therefore, there was no significant interaction effect in muscle stiffness between the BFR and non-BFR groups.

Radiographic outcomes. There were no significant differences in palmar tilt and radial inclination before intervention between the BFR and non-BFR groups, indicating that the severity of the participants' Colles' fractures did not significantly differ between the two groups. There were no significant differences in palmar tilt and radial inclination after intervention between the two groups, indicating that BFR training was safe for patients with Colles' fracture.

Limitations. There were several limitations to our study. First, participants could not be blinded to the use of BFR training, and this may have resulted in participants experiencing a placebo effect. Furthermore, there may be confounding variables at play that could alter our results, including participant motivation and secondary gain.

CONCLUSIONS

This study found that, compared with the traditional rehabilitation, individualized BFR training combined with traditional rehabilitation resulted in a larger increase in PRWE score, grip strength, and ROM of wrist (ulnar deviation) in participants who had undergone plaster cast treatment of Colles' fracture, which meant that participants could engage in their work better. Therefore, individualized BFR combined with traditional rehabilitation might be a better option for rehabilitation of patients of this kind.

The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

The authors declare no potential conflicts of interest with respect to the research, authorship, and/or publication of this article. There are no associations such as consultancies, equity interests, or patent-licensing arrangements.

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